

Siteng Huang

siteng.huang@gmail.com · kyonhuang.top · Google Scholar (2000+ citations) · Github (5k+ stars)

Work Experience

DAMO Academy, Alibaba Group

Algorithm Expert

Hangzhou, China

2024/09 – Present

- Embodied Intelligence Lab for RynnBot
- Projects: RynnVLA-001, RynnVLA-002, RynnBrain, WorldVLA, AffordDex

TongYi Lab & DAMO Academy, Alibaba Group

Research Intern

Hangzhou, China

2022/03 – 2024/07

- Fundamental Visual Intelligence Team for Tongyi Wanxiang (Wan)
- Directors: Gong Biao, Yu Liu, and Deli Zhao

Education

Zhejiang University

Ph.D. - Computer Science

Hangzhou, China

2019/09 – 2024/06

- Thesis: Model Transfer for Multimodal Understanding and Generation

Westlake University

Visiting Student & Joint Program

Hangzhou, China

2018/10 – 2024/06

- Affiliated with Machine Intelligence Laboratory (MiLAB), advisor: Prof. Donglin Wang

Wuhan University

Bachelor of Engineering - Software Engineering

Wuhan, China

2015/09 – 2019/06

- GPA: 3.7/4.0, Rank: 3/244

Peer-reviewed Conference Publications (*: equal contribution, ✉: corresponding author.)

- [C1] **HiF-VLA: Hindsight, Insight and Foresight for Vision-Language-Action models**
Minghui Lin, Pengxiang Ding, Shu Wang, Zifeng Zhuang, Yang Liu, Xinyang Tong, Wenxuan Song, Shangke Lyu, **Siteng Huang**[✉], Donglin Wang[✉]
The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026 (CVPR 2026)
- [C2] **Variation-aware Vision Token Dropping for Faster Large Vision-Language Models**
Junjie Chen, Xuyang Liu, Zichen Wen, Yiyu Wang, **Siteng Huang**, Honggang Chen
The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026 (CVPR 2026)
- [C3] **RynnVLA-001: Using Human Demonstrations to Improve Robot Manipulation**
Yuming Jiang, **Siteng Huang**, Shengke Xue, Yaxi Zhao, Jun Cen, Sicong Leng, Kehan Li, Jiayan Guo, Kexiang Wang, Mingxiu Chen, Fan Wang, Deli Zhao, Xin Li
The 2026 IEEE International Conference on Robotics and Automation (ICRA 2026)
- [C4] **VLA-Adapter: An Effective Paradigm for Tiny-Scale Vision-Language-Action Model**
Yihao Wang*, Pengxiang Ding*, Lingxiao Li, Can Cui, Zirui Ge, Xinyang Tong, Wenxuan Song, Han Zhao, Wei Zhao, Pengxu Hou, **Siteng Huang**, Yifan Tang, Wenhui Wang, Ru Zhang, Jianyi Liu, Donglin Wang
The 40th AAAI Conference on Artificial Intelligence (AAAI 2026)
- [C5] **Towards Affordance-Aware Robotic Dexterous Grasping with Human-like Priors**
Haoyu Zhao*, Linghao Zhuang*, Xingyue Zhao*, Cheng Zeng, Haoran Xu, Yuming Jiang, Jun Cen, Kexiang Wang, Jiayan Guo, **Siteng Huang**[✉], Xin Li, Deli Zhao, Hua Zou[✉]
The 40th AAAI Conference on Artificial Intelligence (AAAI 2026)
- [C6] **Filter, Correlate, Compress: Training-Free Token Reduction for MLLM Acceleration**
Yuhang Han*, Xuyang Liu*, Zihan Zhang, Pengxiang Ding, Junjie Chen, Honggang Chen, Donglin Wang, Qingsen Yan, **Siteng Huang**[✉]
The 40th AAAI Conference on Artificial Intelligence (AAAI 2026)

- [C7] **Global Compression Commander: Plug-and-Play Inference Acceleration for High-Resolution Large Vision-Language Models**
Xuyang Liu, Ziming Wang, Junjie Chen, Yuhang Han, Yingyao Wang, Jiale Yuan, Jun Song, **Siteng Huang**, Honggang Chen
The 40th AAAI Conference on Artificial Intelligence (AAAI 2026)
- [C8] **SSR: Enhancing Depth Perception in Vision-Language Models via Rationale-Guided Spatial Reasoning**
Yang Liu, Ming Ma, Xiaomin Yu, Pengxiang Ding, Han Zhao, Mingyang Sun, **Siteng Huang**, Donglin Wang
The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025)
- [C9] **Long-VLA: Unleashing Long-Horizon Capability of Vision Language Action Model for Robot Manipulation**
Yiguo Fan, Shuanghao Bai, Xinyang Tong, Pengxiang Ding, Yuyang Zhu, Hongchao Lu, Fengqi Dai, Wei Zhao, Yang Liu, **Siteng Huang**, Zhaoxin Fan, Badong Chen, Donglin Wang
The 9th Annual Conference on Robot Learning (CoRL 2025)
- [C10] **CARP: Visuomotor Policy Learning via Coarse-to-Fine Autoregressive Prediction**
Zhefei Gong, Pengxiang Ding, Shangke Lyu, **Siteng Huang**, Mingyang Sun, Wei Zhao, Zhaoxin Fan, Donglin Wang
International Conference on Computer Vision 2025 (ICCV 2025)
- [C11] **QUART-Online: Latency-Free Large Multimodal Language Model for Quadruped Robot Learning**
Xinyang Tong, Pengxiang Ding, Donglin Wang, Wenjie Zhang, Can Cui, Mingyang Sun, Yiguo Fan, Han Zhao, Hongyin Zhang, Yonghao Dang, **Siteng Huang**, Shangke Lyu
The 2025 IEEE International Conference on Robotics and Automation (ICRA 2025)
- [C12] **Accelerating Diffusion Transformers with Token-wise Feature Caching**
Chang Zou*, Xuyang Liu*, Ting Liu, **Siteng Huang**, Linfeng Zhang
The 13th International Conference on Learning Representations (ICLR 2025)
- [C13] **Cobra: Extending Mamba to Multi-Modal Large Language Model for Efficient Inference**
Han Zhao, Min Zhang, Wei Zhao, Pengxiang Ding, **Siteng Huang**, Donglin Wang
The 39th AAAI Conference on Artificial Intelligence (AAAI 2025)
- [C14] **ProFD: Prompt-Guided Feature Disentangling for Occluded Person Re-Identification**
Can Cui*, **Siteng Huang***, Wenxuan Song, Pengxiang Ding, Zhang Min, Donglin Wang
ACM Multimedia 2024 (ACMMM 2024)
- [C15] **PiTe: Pixel-Temporal Alignment for Large Video-Language Model**
Yang Liu, Pengxiang Ding, **Siteng Huang**, Min Zhang, Han Zhao, Donglin Wang
European Conference on Computer Vision 2024 (ECCV 2024)
- [C16] **QUAR-VLA: Vision-Language-Action Model for Quadruped Robots**
Pengxiang Ding, Han Zhao, Wenxuan Song, Wenjie Zhang, Min Zhang, **Siteng Huang**, Ningxi Yang, *et al.*
European Conference on Computer Vision 2024 (ECCV 2024)
- [C17] **Learning Disentangled Identifiers for Action-Customized Text-to-Image Generation**
Siteng Huang, Biao Gong, Yutong Feng, Xi Chen, Yuqian Fu, Yu Liu, Donglin Wang
IEEE/CVF Conference on Computer Vision and Pattern Recognition 2024 (CVPR 2024)
- [C18] **Troika: Multi-Path Cross-Modal Traction for Compositional Zero-Shot Learning**
Siteng Huang, Biao Gong, Yutong Feng, Yiliang Lv, Donglin Wang
IEEE/CVF Conference on Computer Vision and Pattern Recognition 2024 (CVPR 2024)
- [C19] **Check, Locate, Rectify: A Training-Free Layout Calibration System for Text-to-Image Generation**
Biao Gong*, **Siteng Huang***, Yutong Feng, Shiwei Zhang, Yuyuan Li, Yu Liu
IEEE/CVF Conference on Computer Vision and Pattern Recognition 2024 (CVPR 2024)
- [C20] **DARA: Domain- and Relation-aware Adapters Make Parameter-efficient Tuning for Visual Grounding**
Ting Liu, Xuyang Liu, **Siteng Huang**, Honggang Chen, Quanjun Yin, Long Qin, Donglin Wang, Yue Hu
IEEE Conference on Multimedia Expo 2024 (ICME 2024)
- [C21] **VGDifZero: Text-to-image Diffusion Models Can Be Zero-shot Visual Grounders**
Xuyang Liu*, **Siteng Huang***, Yachen Kang, Honggang Chen, Donglin Wang
IEEE International Conference on Acoustics, Speech and Signal Processing 2024 (ICASSP 2024)

- [C22] **Prompt-based Distribution Alignment for Unsupervised Domain Adaptation**
Shuanghao Bai, Min Zhang, Wanqi Zhou, **Siteng Huang**, Zhirong Luan, Donglin Wang, Badong Chen
The 38th AAAI Conference on Artificial Intelligence (AAAI 2024)
- [C23] **VoP: Text-Video Co-operative Prompt Tuning for Cross-Modal Retrieval**
Siteng Huang, Biao Gong, Yulin Pan, Jianwen Jiang, Yiliang Lv, Yuyuan Li, Donglin Wang
IEEE/CVF Conference on Computer Vision and Pattern Recognition 2023 (CVPR 2023)
- [C24] **Reference-Limited Compositional Zero-Shot Learning**
Siteng Huang, Qiyao Wei, Donglin Wang
ACM International Conference on Multimedia Retrieval 2023 (ICMR 2023)
- [C25] **Tree Structure-Aware Few-Shot Image Classification via Hierarchical Aggregation**
Min Zhang, **Siteng Huang**, Wenbin Li, Donglin Wang
European Conference on Computer Vision 2022 (ECCV 2022)
- [C26] **Domain Generalized Few-shot Image Classification via Meta Regularization Network**
Min Zhang, **Siteng Huang**, Donglin Wang
IEEE International Conference on Acoustics, Speech and Signal Processing 2022 (ICASSP 2022)
- [C27] **HINFSHOT: A Challenge Dataset for Few-Shot Node Classification in Heterogeneous Information Network**
Zifeng Zhuang, Xintao Xiang, **Siteng Huang**, Donglin Wang
ACM International Conference on Multimedia Retrieval 2021 (ICMR 2021)
- [C28] **Pareto Self-Supervised Training for Few-Shot Learning**
Zhengyu Chen, Jixie Ge, Heshen Zhan, **Siteng Huang**, Donglin Wang
IEEE/CVF Conference on Computer Vision and Pattern Recognition 2021 (CVPR 2021)
- [C29] **Attributes-Guided and Pure-Visual Attention Alignment for Few-Shot Recognition**
Siteng Huang, Min Zhang, Yachen Kang, Donglin Wang
The 35th AAAI Conference on Artificial Intelligence (AAAI 2021)
- [C30] **DSANet: Dual Self-Attention Network for Multivariate Time Series Forecasting**
Siteng Huang, Donglin Wang, Xuehan Wu, Ao Tang
The 28th ACM International Conference on Information and Knowledge Management (CIKM 2019)

Peer-reviewed Journal Publications (*: equal contribution, ✉: corresponding author.)

- [J1] **High-Fidelity Simulated Data Generation for Real-World Zero-Shot Robotic Manipulation Learning with Gaussian Splatting**
Haoyu Zhao*, Cheng Zeng*, Linghao Zhuang*, Yaxi Zhao, Shengke Xue, Hao Wang, Xingyue Zhao, Zhongyu Li, Kehan Li, **Siteng Huang**✉, Mingxiu Chen, Xin Li, Deli Zhao, Hua Zou✉
IEEE Robotics and Automation Letters (RA-L 2026)
- [J2] **M2IST: Multi-Modal Interactive Side-Tuning for Memory-efficient Referring Expression Comprehension**
Xuyang Liu, Ting Liu, **Siteng Huang**✉, Yue Hu, Qianjun Yin, Donglin Wang, Honggang Chen✉
IEEE Transactions on Circuits and Systems for Video Technology (TCSVT 2025)

Preprints & Under Review (Only including publicly available work.)

- [P1] **RynnBrain: Open Embodied Foundation Models**
Ronghao Dang, Jiayan Guo, Bohan Hou, Sicong Leng, Kehan Li, Xin Li, Jiangpin Liu, Yunxuan Mao, Zhikai Wang, Yuqian Yuan, Minghao Zhu, Xiao Lin, Yang Bai, Qian Jiang, Yaxi Zhao, Minghua Zeng, Junlong Gao, Yuming Jiang, Jun Cen, **Siteng Huang**, Liuyi Wang, Wenqiao Zhang, Chengju Liu, Jianfei Yang, Shijian Lu, Deli Zhao
- [P2] **MMaDA-VLA: Large Diffusion Vision-Language-Action Model with Unified Multi-Modal Instruction and Generation**
Yang Liu, Pengxiang Ding, Tengyue Jiang, Xudong Wang, Wenxuan Song, Minghui Lin, Han Zhao, Hongyin Zhang, Zifeng Zhuang, Wei Zhao, **Siteng Huang**, Jinkui Shi, Donglin Wang

- [P3] **Articulat3D: Reconstructing Articulated Digital Twins From Monocular Videos with Geometric and Motion Constraints**
Lijun Guo, Haoyu Zhao, Xingyue Zhao, Rong Fu, Linghao Zhuang, **Siteng Huang**, Zhongyu Li, Hua Zou
- [P4] **RynnVLA-002: A Unified Vision-Language-Action and World Model**
Jun Cen*, **Siteng Huang***, Yuqian Yuan*, Kehan Li*, Hangjie Yuan, Chaohui Yu, Yuming Jiang, Jiayan Guo, Hao Luo, Fan Wang, Xin Li, Deli Zhao, Hao Chen
- [P5] **WorldVLA: Towards Autoregressive Action World Model**
Jun Cen, Chaohui Yu, Hangjie Yuan, Yuming Jiang, **Siteng Huang**, Jiayan Guo, Xin Li, Yibing Song, Hao Luo, Fan Wang, Deli Zhao, Hao Chen
- [P6] **VARD: Efficient and Dense Fine-Tuning for Diffusion Models with Value-based RL**
Fengyuan Dai*, Zifeng Zhuang*, Yufei Huang, **Siteng Huang**, Bangyan Liao, Donglin Wang, Fajie Yuan
- [P7] **SSR: Enhancing Depth Perception in Vision-Language Models via Rationale-Guided Spatial Reasoning**
Yang Liu, Ming Ma, Xiaomin Yu, Pengxiang Ding, Han Zhao, Mingyang Sun, **Siteng Huang**, Donglin Wang
- [P8] **OpenHelix: A Short Survey, Empirical Analysis, and Open-Source Dual-System VLA Model for Robotic Manipulation**
Can Cui, Pengxiang Ding, Wenxuan Song, Shuanghao Bai, Xinyang Tong, Zirui Ge, Runze Suo, Wanqi Zhou, Yang Liu, Bofang Jia, Han Zhao, **Siteng Huang**, Donglin Wang
- [P9] **Unicorn: Text-Only Data Synthesis for Vision Language Model Training**
Xiaomin Yu, Pengxiang Ding, Wenjie Zhang, **Siteng Huang**, Songyang Gao, Chengwei Qin, Kejian Wu, Zhaoxin Fan, Ziyue Qiao, Donglin Wang
- [P10] **Exploring the Evolution of Physics Cognition in Video Generation: A Survey**
Minghui Lin, Xiang Wang, Yishan Wang, Shu Wang, Fengqi Dai, Pengxiang Ding, Cunxiang Wang, Zhen-grong Zuo, Nong Sang, **Siteng Huang**[✉], Donglin Wang[✉]
- [P11] **Humanoid-VLA: Towards Universal Humanoid Control with Visual Integration**
Pengxiang Ding, Jianfei Ma, Xinyang Tong, Binghong Zou, Xinxin Luo, Yiguo Fan, Ting Wang, Hongchao Lu, Panzhong Mo, Jinxin Liu, Yuefan Wang, Huaicheng Zhou, Wenshuo Feng, Jiacheng Liu, **Siteng Huang**, Donglin Wang
- [P12] **Score and Distribution Matching Policy: Advanced Accelerated Visuomotor Policies via Matched Distillation**
Bofang Jia, Pengxiang Ding, Can Cui, Mingyang Sun, Pengfang Qian, **Siteng Huang**, Zhaoxin Fan, Donglin Wang
- [P13] **Sparse-Tuning: Adapting Vision Transformers with Efficient Fine-tuning and Inference**
Ting Liu, Xuyang Liu, **Siteng Huang**, Liangtao Shi, Zunnan Xu, Yi Xin, Qunjun Yin, Xiaohong Liu
- [P14] **Focus-Consistent Multi-Level Aggregation for Compositional Zero-Shot Learning**
Fengyuan Dai, **Siteng Huang**, Min Zhang, Biao Gong, Donglin Wang

Professional Services

- **Journal reviewer** for RA-L, TNNLS, TCSVT, TMM, ACM TIST, ACM TOIS, JVC, CPE
- **Conference reviewer** for CVPR, ICCV, ECCV, NeurIPS, AAAI, IJCAI, ACMMM, CoRL, ICRA, ICME, ICMR, ACCV, ICPR
- **Session chair** for AAAI 2026, the First Westlake Robot Learning Symposium

Selected Talks

- **Physical AI Ecosystem: Tackling the Key Barriers to Embodied Intelligence**
AAAI 2026 Interactive Industry Sessions 2026/01
- **Perception, Reasoning, and Action: Empowering Robots with Embodied Foundation Models**
AAAI 2026 Alibaba Booth Talk 2026/01
- **RynnVLA Series Embodied Manipulation Large Model**
Tsinghua University (2025/12), Hunan University (2025/12), XJTU (2026/03) 2025-2026
- **Core Engine for General Embodied Intelligence: World Models - Current Status, Progress and Roadmap**
MIIT 2025/10